Statistical analysis of geochemical Datasets using PCA Approach

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CENTRAL UNIVERSITY OF KARNATAKA

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2021

DEPARTMENT OF GEOLOGY
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CERTIFICATE

This is to certify that the internship entitled **"Statistical analysis of geochemical data sets using PCA approach**" is a bonafide record of the research work carried out by **Ms Deepshikha Srivastava**, at National Centre for Earth Science Studies, Akkulam, Thiruvananthapuram (during May-June 2021), under the guidance of Dr. Chandra Prakash Dubey, Scientist C, Solid Earth Research Group (SERG), NCESS, Thiruvananthapuram for the partial fulfillment of the requirement for the award of the degree of Master of Science in Applied geology , Central University of Karnataka.

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June 2021

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DECLARATION

I, DEEPSHIKHA SRIVASTAVA hereby declare that the dissertation entitled **"Statistical analysis of geochemical data sets using PCA approach"** is submitted to Central University of Karnataka in partial fulfillment of requirements for the award of master degree in Geology. This is a record of original work done by me under the supervision of Dr. Chandra Prakash Dubey, Scientist C, Solid Earth Research Group (SERG), ESSO-National Centre for Earth Science Studies, Thiruvananthapuram and is not been submitted elsewhere for any other degree.

Deepshikha

Place: Kadaganchi DEEPSHIKHA SRIVASTAVA

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16-05-2021 **DEEPSHIKHA SRIVASTAVA**

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1.1 Introduction

Geostatistical tool is a set of statistical tools that describes the temporal and/or spatial distribution of a given area of interest; about which one only knows its value at discrete and sparse locations (Goovaerts 1997). To the study of spatially distributed data this is also defined as the application of theory of regionalized variable. The significance of geostatistical tools has grown significantly in geoscience due to their ability to integrate, within the same framework for example, in the reservoir grid of geophysical data and well-log data of a diverse nature and support. In geoscience study the datasets collected and analyzed, includes many individual observations and all these types of data produces dataset of various dimension, which is very difficult for a human to visualize and interpret. To overcome this difficulty many of the geoscientist applies the geostatistical tools to their dataset of multi-dimensional data. The geostatistical tools help to study the variance and correlation between the variables and to evaluate the differences and similarity among the dataset.

There are two types of geostatistical methods: Bivariate statistical analysis and multivariate statistical analysis. Bivariate statistical analysis means the analysis of datasets having two variables and if the datasets have more than two variables we use multivariate statistical analysis. Constantin et al., (2011) used bivariate statistical method to do the landslide susceptibility assessment on the basis of comparison between the landslide inventory map as a dependent variable and the single input parametric maps (lithology/landslides, slope angle/landslides, etc.). There are several multivariate analysis used for the visualizing the geochemical data, such as Principal Component Analysis (PCA), Discriminant Analysis, Linear Discriminant Analysis (LDA), Factor Analysis, Singular Value Decomposition and Hierarchical Cluster Analysis (HCA). Multivariate statistical analysis has been used successfully in a number of geochemical studies such as for analyzing geochemical data of stream sediment, soil or estuarine sediments to identify mineralization or contamination over natural geochemical background (Chork and Salminen, 1993). Multivariate statistical analysis has also been used in Petrology to identify diagenetic processes and effect of provenance on the bulk chemistry of rock or sediment (Hakstege et al., 1993).

The hierarchical cluster analysis (HCA) and geostatistical analysis are useful tools for the identification of pollution sources (Li and Feng, 2012; Lu et al., 2012). In hydro-geochemical studies also (Cameron, 1996) multivariate statistical analysis have been used widely to recognize a range of processes that control water chemistry, including seasonal variation in surface water composition, recharge area, salt water intrusion in fresh water aquifers, ground water contamination, and natural mineral dissolution. Helvoort et al., (2005) used multivariate statistical analysis of heterogeneous geochemical datasets for the bulk chemical characterization of fluvial deposits. Garcia et al., (2020) demonstrated the usefulness of the multivariate statistical approach in geochemical parameters to distinguish depositional paleo-environments from source rocks. Reid and Spencer (2009) demonstrated use of principal components analysis (PCA) on

estuarine sediment datasets in the effect of data pre-treatment. Stokes and Lowe (1988) used LDA and the major element chemistry of glass shards, particularly iron oxide, to separate groups of volcanic epicenters in New Zealand.

Principal components analysis (PCA) is a multivariate statistical exploratory technique that has been used to examine data variability. It is often applied to the datasets where it is large and difficult to interpret, and where the variables are complex inter-relationships and are difficult to identify and visualize. PCA is a multivariate statistical method used for reducing a large set of possibly associated variables to a smaller number of uncorrelated variables known as principal components which contains valuable information of the original dataset (Wishart, 2013). A principal component (PC) factor is a linear combination of the original variables that is weighted by their contribution to explaining the variance in a particular orthogonal dimension (Geladi et al., 1989; Geladi and Grahn, 1996). The first Principal Component (PC1) elucidates the greatest possible portion of the variance within the dataset, whereas the second Principal Component (PC2) captures the second largest variance orthogonal to the first, and so on. Since patterns in data can be hard to find in data of high dimension, where the luxury of graphical representation is not available, PCA is a powerful tool for analyzing data. PCA is very useful in geological data especially in sedimentology (Webb and Briggs, 1966). PCA transforms the data of n-dimension into data along axes of great variance. The first principal component is parallel to the axis of greatest variance in the matrix; the second principal component is orthogonal to the first and so on.

Thus, present project aims to identify the geochemical variability in the study area by using the well-proven multivariate method to analyze the geochemical data, the principal components analysis (PCA). First we will try to implement PCA over randomly generated datasets followed by real geological example from field area of Nashville Dome, United States.

1.2 Computational Methodology

PCA is a mathematical procedure to transform a large number of correlated variables into smaller number of uncorrelated variables known as Principle Component. For doing PCA of any given matrix of n variables and m samples, we first center the data on the means of each variable, this will center the data on the origin of our principal components and this process is known as adjusting the data.

For mean we use the formula;

$$\text{Mean} = \Sigma^{n}_{i=1} Xi/n \qquad \text{(Eq. 1)}$$

For adjusting data we use the formula;

$$\Sigma^{n}_{i=1} Xi - Xm \qquad \text{(Eq. 2)}$$

where, Xi is the ith element of the variable X and Xm is the mean of X variables;

n is the number of elements present.

We then calculate the covariance matrix of the data which tells about the spatial relationship and variance of the datasets with respect to different variables.

For calculating variance we use the formula;

$$Var(X,X)=\Sigma^{n}_{i=1}\frac{(Xi-Xm)^2}{(n-1)} \qquad \text{(Eq. 3)}$$

For calculating covariance we use the formula;

$$Cov(X,Y)=\Sigma^{n}_{i=1}\frac{(Xi-Xm)\,(Yi-Ym)}{(n-1)} \qquad \text{(Eq. 4)}$$

where, Xi is the ith element of the variable X and Xm is the mean of X variables;
Yi is the ith element of the variable Y and Ym is the mean of Y variables and
n is the number of variables present in any of the dimension.

From covariance matrix we can calculate the Eigenvalues which is also referred as scores, and Eigenvectors which is also referred as loadings. The eigenvectors and eigenvalues always come in pairs, the eigenvalues determines the magnitude and eigenvectors determines the direction of data with greatest variance in the dataset. The eigenvector with the highest eigenvalue is the first principal component and accounts for the greatest possible variance in the dataset.

For calculating eigenvalues and eigenvectors we use the formula;

$$\det(A - \lambda I)X=0 \qquad \text{(Eq. 5)}$$

where, A is a covariance matrix,
I is identity matrix,
λ is the eigenvalue and
X is eigenvector matrix.

For plotting the final plot we choose the eigenvectors of higher significance and refer it as feature vector. At last we multiply the transformed feature vector into the transformed adjusted data to get maximum information about original data.

For calculating final data we use the formula;

Final Data = Row Feature Vector x Row Data Adjust (Eq. 6)

where, Row feature vector is the eigenvectors transposed and
Row data adjust is the mean adjusted data of the original data transposed.

1.3 Matlab Algorithm Development

MATLAB is a programming language and numeric computing environment developed by MathWorks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, etc. This was designed by Cleve Moler and the MATLAB we used for our study is R2018a version.

When MATLAB is started, the desktop appears in its default layout:

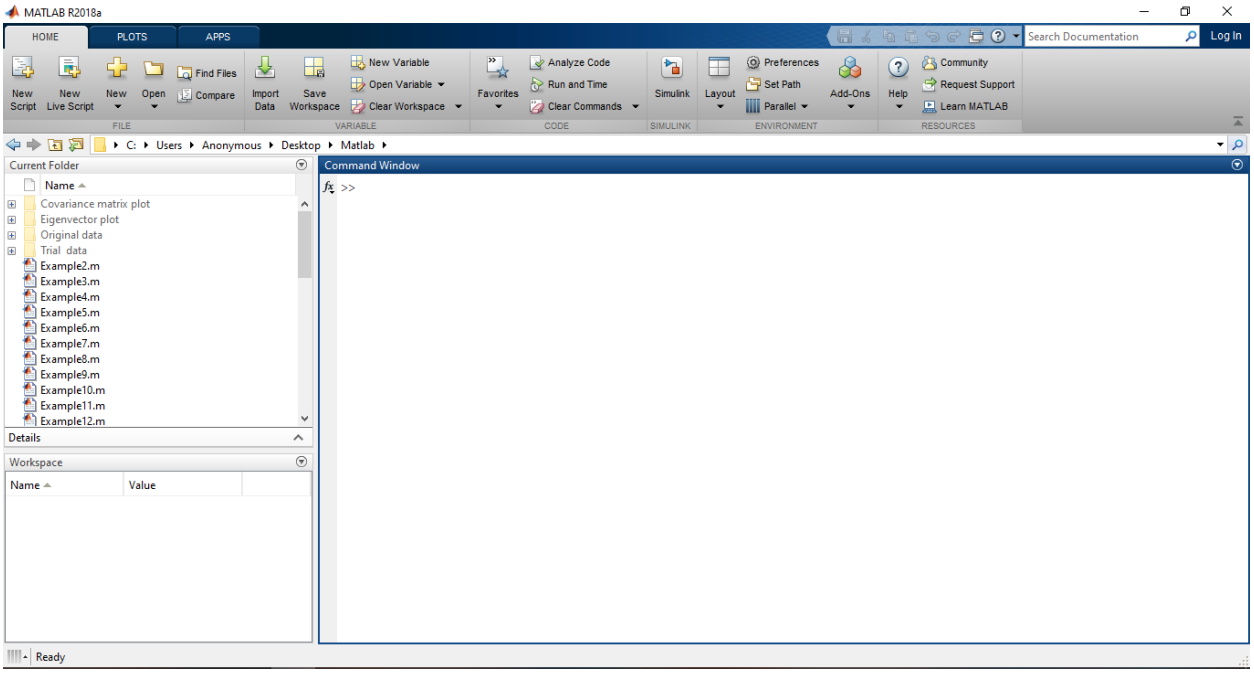

Figure 1. The Default Desktop Panel of Matlab

The desktop has the following panels:

Current Folder - This panel allows you to access the project folders and files.

Command Window - This is the main area where commands can be entered at the command line. It is indicated by the command prompt (>>).

Workspace - The workspace shows all the variables created and/or imported from files.

Command History - This panel shows or rerun commands that are entered at the command line. MATLAB environment behaves like a super-complex calculator. You can enter commands at the >> command prompt.

Scripts files- These files are program files with .m extension. In these files, you write series of commands, which you want to execute together.

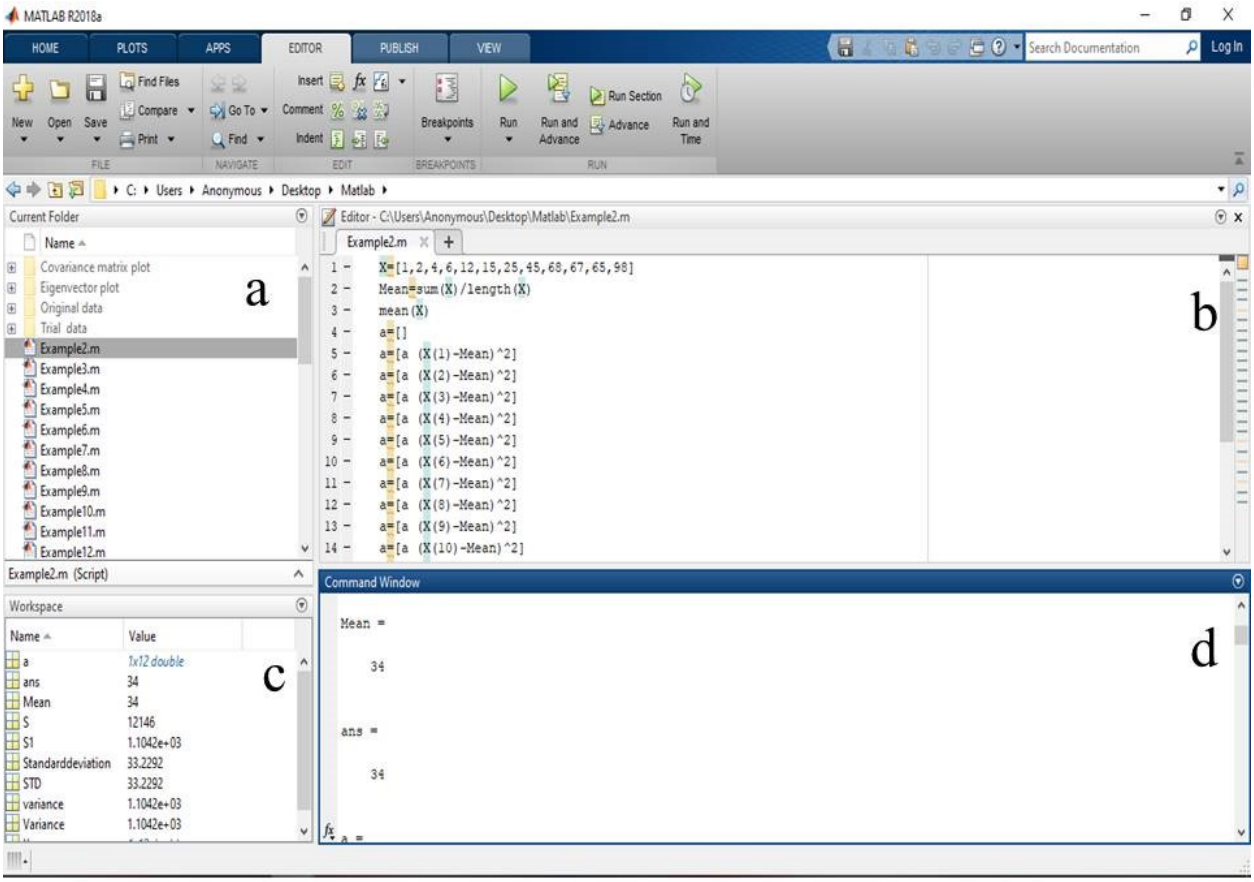

Figure 2. Desktop Panel of Matlab with any file in use, a) Current folder, b) Script file, c) Workspace and d) Command Window.

In command window if we type any expression and when we press enter we get the answer:

For example in command window when we type

```
>>10+10
ans =
        20
```

Semicolon (;) indicates end of statement. However, if you want to suppress and hide the MATLAB output for an expression, add a semicolon after the expression. The percent symbol (%) is used for indicating a comment line.

MATLAB supports the following commonly used operators and special characters: **'+'** Plus or addition operator; **'-'** Minus or subtraction operator; **'*'** Scalar and matrix multiplication operator; **'.*'** Array multiplication operator; **'^'** Scalar and matrix exponentiation operator; **'.^'** Array exponentiation operator; **'\'** Left-division operator; **'/'** Right-division operator; **'.\'** Array left-division operator; **'./'** Array right-division operator; **':'** Colon or generates regularly spaced elements and represents an entire row or column; **'( )'** Parentheses or encloses function

arguments and array indices or overrides precedence; **'[ ]'** Brackets or enclosures array elements; **'.'** Decimal point **'…'** Ellipsis; line-continuation operator; **','** Comma or separates statements and elements in a row; **';'** Semicolon; separates columns and suppresses display; **'%'** Percent sign or designates a comment and specifies formatting; **'_'** Quote sign and transpose operator; **'._'** Non-conjugated transpose operator; **'='** Assignment operator.

MATLAB supports the following special variables and constants: **'ans'** Most recent answer; **'eps'** Accuracy of floating-point precision; **'i,j'** The imaginary unit √-1. Inf Infinity; **'NaN'** Undefined numerical result (not a number); **'pi'** The number π.

Variable names consist of a letter followed by any number of letters, digits or underscore. MATLAB is case-sensitive. Variable names can be of any length, however, MATLAB uses only first N characters, where N is given by the function namelengthmax.

The save command is used for saving all the variables in the workspace, as a file with .mat extension, in the current directory. For example, you can type 'save myfile' in the script and you can reload the file anytime later using the load command such as type 'load myfile' in the script.

In MATLAB environment, every variable is an array or matrix. You can assign variables in a simple way. For example, if you type x = 3, it creates a 1-by-1 matrix named x and stores the value 3 in its element.

A vector is a one-dimensional array of numbers. MATLAB allows creating two types of vectors: Row vectors and Column vectors Row vectors are created by enclosing the set of elements in square brackets, using space or comma to delimit the elements. For example, A = [5,6,7,8] which gives the result as;

A = 5 6 7 8

Column vectors are created by enclosing the set of elements in square brackets, using semicolon (;) to delimit the elements. B = [1; 2; 3; 14] which gives the result as;

```
B=1
   2
   3
   14
```

In MATLAB, a matrix is created by entering each row as a sequence of space or comma separated elements, and end of a row is demarcated by a semicolon. For example, let us create a 2-by-2 matrix as: C = [1 2 ; 4 5]

```
C =  1   2
     4   5
```

The other commands commonly used are: **'clc'** clears command window; **'clear'** removes variables from memory; **'length'** computes number of elements; **'max'** returns largest element; **'min'** returns smallest element; **'size'** Computes array size; **'sort'** Sorts each column; **'sum'** Sums each column; **'eye'** Creates an identity matrix; **'ones'** Creates an array of ones; **'zeros'** Creates an array of zeros; **'det'** Computes determinant of an array; **'inv'** Computes inverse of a matrix; **'pinv'** Computes pseudoinverse of a matrix; **'plot'** Generates xy plot; **'print'** Prints plot or saves plot to a file; **'title'** Puts text at top of plot; **'xlabel'** Adds text label to x-axis; **'ylabel'** Adds text label to y-axis; **'axes'** Creates axes objects; **'close'** Closes the current plot; **'close all'** Closes all plots; **'figure'** Opens a new figure window; **'hold'** Freezes current plot; **'legend'** Legend placement by mouse; **'solve'** solves the numerical data; **'double'** double precision numerical data; **'mean'** computes mean of any array or dataset; **'std'** computes standard deviation of an array or dataset; **'cov'** computes covariance matrix of an array; **'eig'** computes eigenvalues of an array and **'Char'** Convert to character array (string).

The following flow chart depicts the steps to do the computational analysis for any given dataset:

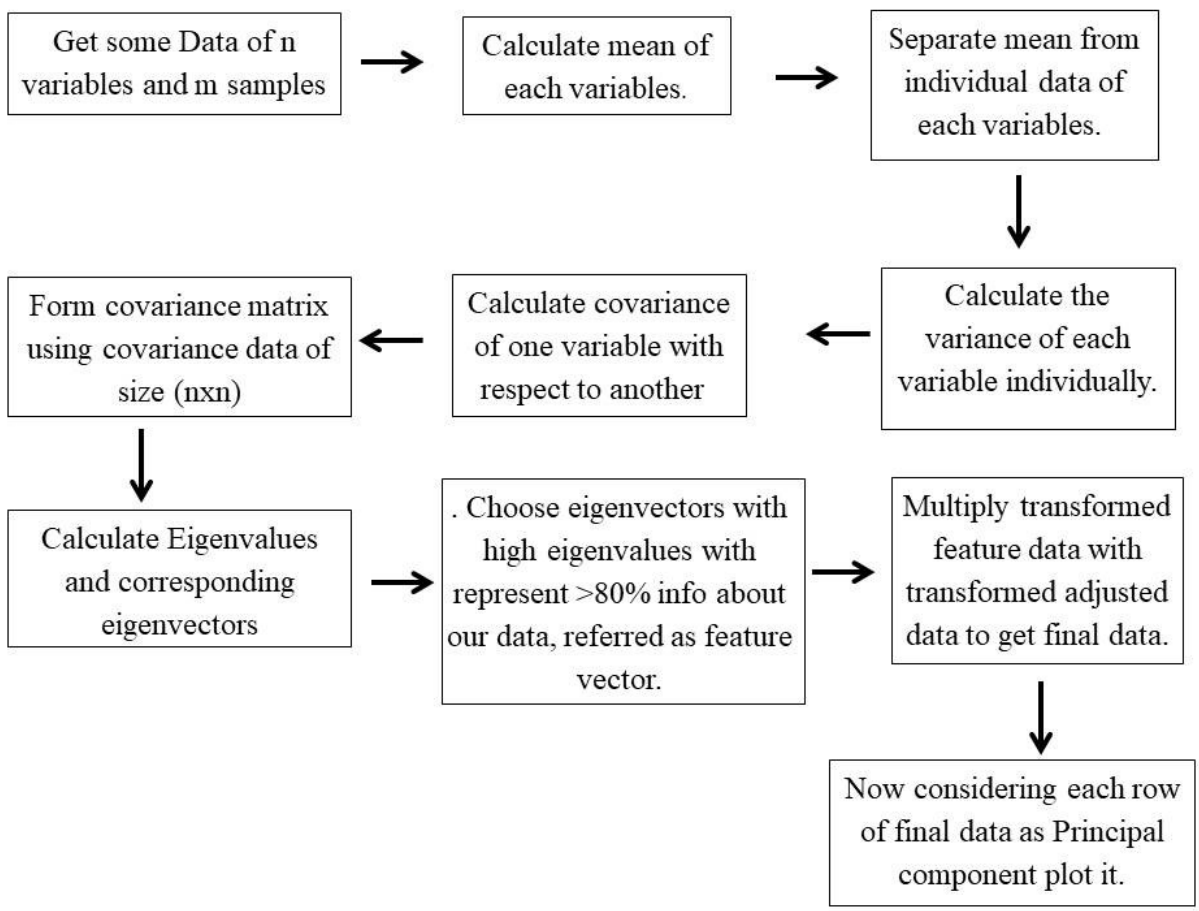

Figure 3. Flow chart representing the steps to do Principal Component Analysis.

Step 1: Get some data

Suppose we take any own made-up dataset of nine school students with marks in five different subjects namely Maths, Physics, English, Chemistry and Hindi and it got five dimensions and dataset is shown below:

Student	Maths	Physics	English	Chemistry	Hindi
S1	80	82	90	85	95
S2	85	85	95	87	93
S3	82	87	92	89	89
S4	78	90	93	86	90
S5	75	84	91	88	94
S6	83	89	94	90	92
S7	84	86	89	92	91
S8	79	91	96	91	96
S9	81	83	87	90	90

Step 2: Subtract the mean

We have to subtract the mean from each of the data dimensions for PCA to work properly as shown in Eq. 2. This is done to normalize or adjust the data or to center the data at zero. The dataset we get after subtracting the mean has mean zero and standard deviation one.

Mean is calculated using formula as shown in Eq. 1.

Adjusted data

Students	Maths	Physics	English	Chemistry	Hindi
S1	-0.78	-4.33	-1.89	-3.67	2.78
S2	4.22	-1.33	3.11	-1.67	0.78
S3	1.22	0.67	0.11	0.33	-3.22
S4	-2.78	3.67	1.11	-2.67	-2.22
S5	-5.78	-2.33	-0.89	-0.67	1.78
S6	2.22	2.67	2.11	1.33	-0.22
S7	3.22	-0.33	-2.89	3.33	-1.22
S8	-1.78	4.67	4.11	2.33	3.78
S9	0.22	-3.33	-4.89	1.33	-2.22

Step 3: Calculate the covariance matrix

The covariance matrix will be a square matrix and the size will be equal to the number of variables present in our data, since we have five dimension data the covariance matrix will be

5x5. We can determine the variance and trend of our data on the basis of covariance matrix, if the non-diagonal elements are positive all the variables either increase together or decrease together but if the non-diagonal elements are negative they are not depended on each other.

For calculating covariance of any element for two dimensional data we use the following formula given in Eq. 4.

The covariance matrix for two dimension data will have five rows and five columns and the values will be as

Covariance matrix

	Maths	Physics	English	Chemistry	Hindi
Maths	9.9444	-0.2917	0.4722	2.1667	-2.3194
Physics	-0.2917	10.0000	6.5417	2.3750	-0.3333
English	0.4722	6.5417	8.6111	-0.5417	2.7778
Chemistry	2.1667	2.3750	-0.5417	5.5000	-0.7917
Hindi	-2.3194	-0.3333	2.7778	-0.7917	5.9444

The diagonal elements tells about the variance of marks within the same subjects whereas non-diagonal elements tells the variance of marks of one subject with another; here we can see that the diagonal elements are positive which indicates that the elements of the same variable increases with each other. The marks of the subjects English and Chemistry increases with increase in Maths marks and Physics and Hindi marks decreases with increase in Maths marks. English and Chemistry marks are positively correlated with Physics marks as well. But English and Chemistry marks are not positively correlated to each other.

Step 4: Calculate the eigenvectors and eigenvalues of the covariance matrix

Next step is to calculate eigenvalues and eigenvector. They are very important because they tell valuable information about our data. For the calculation of eigenvalues and eigenvectors we use the formula given in Eq. 5. We get five eigenvalues which is equal to the number of variables present in raw data and the eigenvector is a square matrix which is of 5x5 size.

Eigenvalues = 0.5732, 4.5110, 6.6418, 11.9860 and 16.2881.

Eigenvector =

	PC5	PC4	PC3	PC2	PC1
Maths	-0.2290	-0.0223	0.5190	0.8232	-0.0027
Physics	-0.5362	-0.1142	-0.4039	0.1001	-0.7255
English	0.6106	-0.1454	0.3990	-0.0878	-0.6626
Chemistry	0.3629	0.7507	-0.3936	0.3691	-0.1163
Hindi	-0.3944	0.6338	0.5033	-0.4103	-0.1451

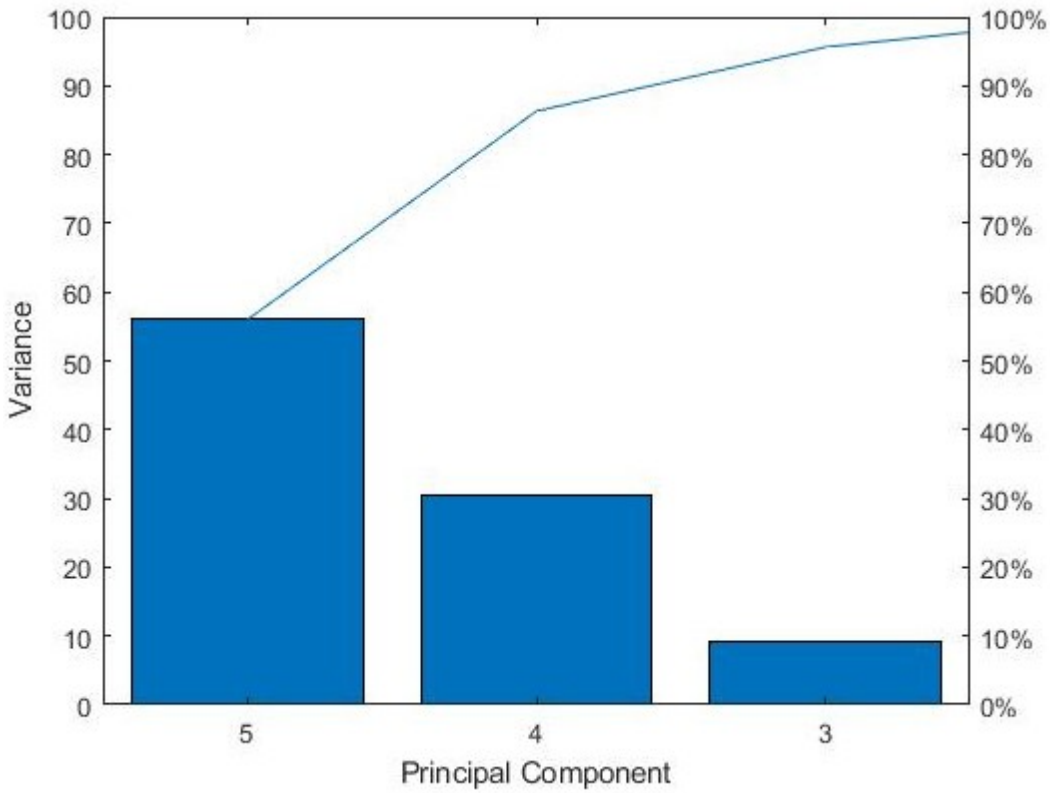

Figure 4. The percent of total variance explained by each principal component.

Step 5: Choosing components and forming a feature vector

The eigenvector and eigenvalues always comes in pair and the eigenvector with the highest eigenvalue is the principle component of the data set. Therefore firstly we will arrange the eigenvalues in descending order and then we will arrange the eigenvector according to their corresponding eigenvalues. We will choose the eigenvalues with higher significance and ignore the eigenvalues with lesser significance and will form a feature vector. Since four of the eigenvalues represents most of the information about our data we will form the feature vector of 5x4 size and will get the feature vector as:

Feature vector =

	PC1	PC2	PC3	PC4
Maths	-0.0027	0.8232	0.5190	-0.0223
Physics	-0.7255	0.1001	-0.4039	-0.1142
English	-0.6626	-0.0878	0.3990	-0.1454
Chemistry	-0.1163	0.3691	-0.3936	0.7507
Hindi	-0.1451	-0.4103	0.5033	0.6338

Step 5: Deriving the new data set

This is the final and most important step for PCA, after we choose the eigenvectors that represents most of our data and form feature vector with it we take the transpose of the feature vector and multiply it with the transposed original mean adjusted data to get the final data as in Eq. 6.

We do this so that we get the original data exclusively in terms of the vectors that we have chosen and we get the transformed data as:

Final data =

4.4209	-1.0245	-0.1318	-2.7563	2.1170	-3.4623	1.9370	-6.9244	5.8245
-3.4013	2.1348	2.5082	-2.0897	-5.8873	2.4942	4.6046	-2.0461	1.6824
3.4342	5.0187	-1.3435	-2.5481	-1.2537	0.2820	-1.2728	-0.1842	-2.1322
-0.2052	-1.1525	-1.9116	-3.9287	1.1508	0.1990	2.1139	3.0549	0.6790

Now considering each row of final data as principal components we get four principal components. Now we can plot all principal components in 2D and 3D plot and interpret it.

1.4 Application of PCA in Geochemical dataset from Nashville Dome, United States

The data used in this study has been taken from the Upper Ordovician carbonate strata exposed on the Nashville Dome, USA. This strata record the development of eleven third-order depositional sequences (Holland and Patzkowsky, 1997, 1998). These sequences are composed of transgressive and highstand systems tracts and have sequence boundaries that display geochemical, petrographic, and, in some outcrops, macroscopic evidence of subaerial exposure and erosion (Holland and Patzkowsky 1998; Railsback et al. 2003) From the western side of the Gladeville road cut two hundred hand samples were collected. Ten vertical transects of samples were collected, yielding 10 replicate samples for each stratigraphic horizon that was sampled. The transect were spaced 2 m apart horizontally. Within each vertical transect, twenty samples were collected below the M4–M5 sequence boundary at 10 cm intervals. Each hand sample was sectioned and polished. The stable isotope analysis and trace element analysis were conducted (Theiling et al., 2007). Figure 5 shows the map of Ordovician outcrop area in the Eastern United States; the figure shows three exposed strata one in the southwest direction in the Nashville dome which is surrounded by Carboniferous strata, second in the center of the map and third strata is lineament trending southwest to northeast. Abbreviations used in the map shows the name of local area.

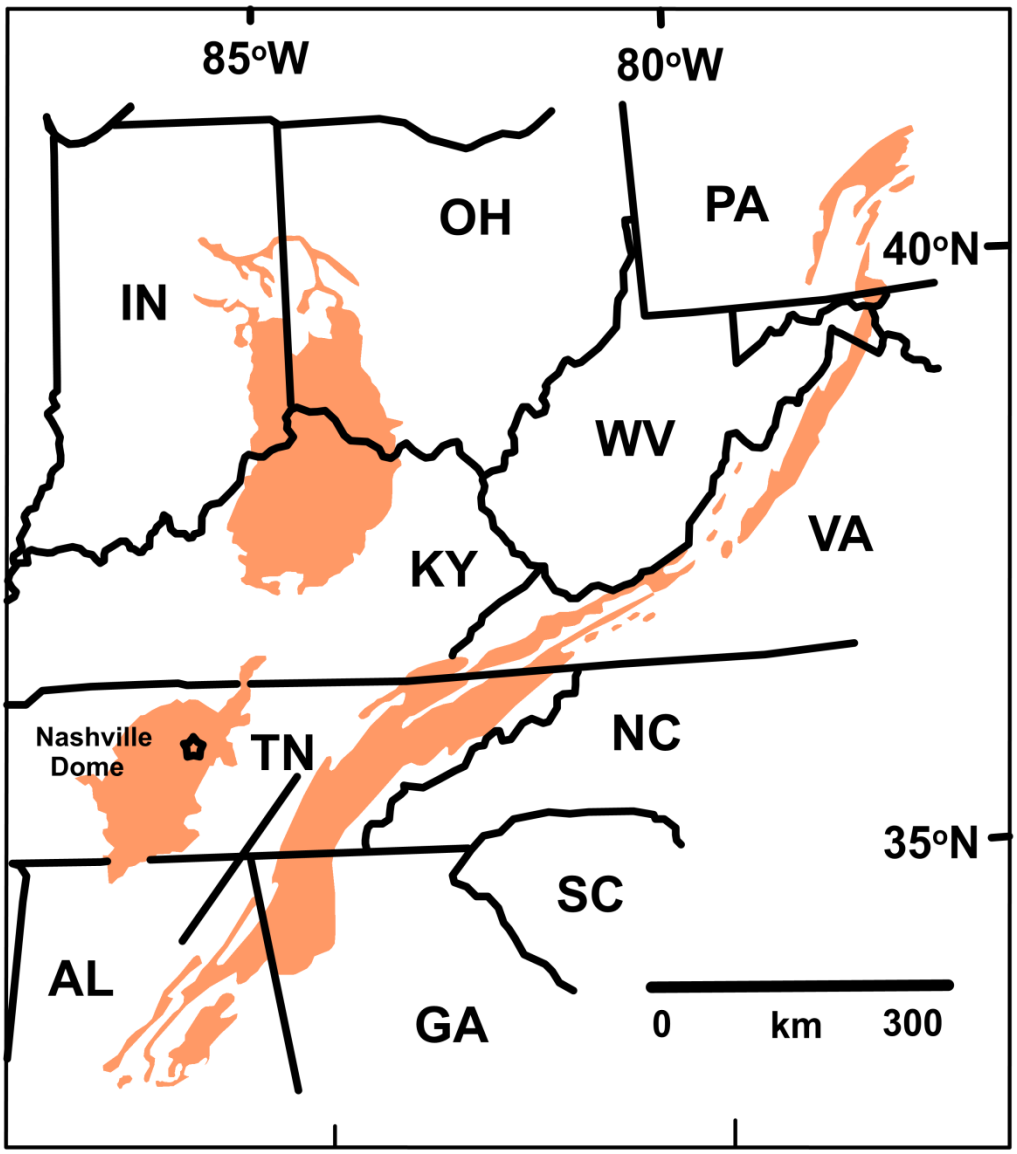

Figure. 5. Map of Ordovician outcrop area (peach) in the eastern United States (Map modified after Theiling et al., 2007). Location of Gladeville sampling locality shown at star. (Abbreviations used: AL=Alabama, GA= Georgia, TN= Tennessee, NC= North Carolina, KN= Kentucky, VA= Virginia, OH= Ohio, SC= South Carolina, WV= West Virginia, PA= Pennsylvania and IN= Indiana).

Firstly we will import the data using import data file present in home tab and then we can save the variable in mat files. Then we will use the following code to get the required data.

```
clc
close all
clear all

% Load Data
```

```
load NashvilleCarbonates.mat
whos
[numobs, numvars] = size(NashvilleCarbonates);


% Variables in the data in NashvilleCarbonates file are presented as
variables=NashvilleCarbonates(1,3:10)

A =NashvilleCarbonates.StratPosition;
B =NashvilleCarbonates.d13C;
C =NashvilleCarbonates.d18O;
D =NashvilleCarbonates.Al;
E =NashvilleCarbonates.Ca;
F =NashvilleCarbonates.Fe;
G =NashvilleCarbonates.Mg;
H =NashvilleCarbonates.Mn;
I =NashvilleCarbonates.Si;

Carbonates=[B,C,D,E,F,G,H,I]


% Overview of the variables
% Let's start off by examining the distributions of each of the variables
figure
boxplot(Carbonates)

xlabel('Carbonate');
ylabel('Value');


% Create standardised data
% We can see from the boxplot that the variances of the limestone impurities
are not equal.
% We would like to be able to consider each variable on the same scale, so
let's standardise
% them to have a mean of 0 and a variance of 1.

means = mean(Carbonates);
sds = std(Carbonates);
metalstd = (Carbonates - repmat(means,numobs,1)) ./ repmat(sds,numobs,1)

figure;
boxplot(metalstd)

xlabel('Carbonate');
ylabel('Values (Standardised)');
legend

xlabel('Carbonate')
ylabel('Values (Standardised)')


% Variables in the data file are presented as
A1 =metalstd(:,1);
```

```
B1 =metalstd(:,2);
C1 =metalstd(:,3);
D1 =metalstd(:,4);
E1 =metalstd(:,5);
F1 =metalstd(:,6);
G1 =metalstd(:,7);
H1 =metalstd(:,8);

% Carry out PCA
% We can extract a lot of information with a principal component analysis
ofthe data.
% PCA constructs new variables (principal components, or PCs) that are
weighted sums
% of the original variables.
% By examining a small number of principal components, you can get an
% understanding of the process which is nearly as good as examining all of
the original
% variables.

% The PCA is done in six steps
% First step is to Standaradize the data

% Mean Caluclation
meanA1= sum(A1)/length(A1);
meanB1= sum(B1)/length(B1);
meanC1= sum(C1)/length(C1);
meanD1= sum(D1)/length(D1);
meanE1= sum(E1)/length(E1);
meanF1= sum(F1)/length(F1);
meanG1= sum(G1)/length(G1);
meanH1= sum(H1)/length(H1);



% Normalise Data
% normalization also known as standardization in PCA
NA=(A1-mean(A1))/std(A1);
NB=(B1-mean(B1))/std(B1);
NC=(C1-mean(C1))/std(C1);
ND=(D1-mean(D1))/std(D1);
NE=(E1-mean(E1))/std(E1);
NF=(F1-mean(F1))/std(F1);
NG=(G1-mean(G1))/std(G1);
NH=(H1-mean(H1))/std(H1);



%The second step is to calculate the covariance matrix
% Covariance Calculation
% calculating the variation of each element from the mean for each column
a1=A1-mean(A1);
b1=B1-mean(B1);
c1=C1-mean(C1);
d1=D1-mean(D1);
e1=E1-mean(E1);
```

```
f1=F1-mean(F1);
g1=G1-mean(G1);
h1=H1-mean(H1);


%Since there are 8 variables the covariance matrix will be of 9x9 size
%calculating the elements of covariance matrix which will be of size(9,9)
Coaa=a1'*a1/(length(A)-1);
Cobb=b1'*b1/(length(B)-1);
Cocc=c1'*c1/(length(C)-1);
Codd=d1'*d1/(length(D)-1);
Coee=e1'*e1/(length(E)-1);
Coff=f1'*f1/(length(F)-1);
Cogg=g1'*g1/(length(G)-1);
Cohh=h1'*h1/(length(H)-1);
Coab=a1'*b1/(length(B)-1);
Coac=a1'*c1/(length(C)-1);
Coad=a1'*d1/(length(D)-1);
Coae=a1'*e1/(length(E)-1);
Coaf=a1'*f1/(length(F)-1);
Coag=a1'*g1/(length(G)-1);
Coah=a1'*h1/(length(H)-1);
Cobc=b1'*c1/(length(C)-1);
Cobd=b1'*d1/(length(D)-1);
Cobe=b1'*e1/(length(E)-1);
Cobf=b1'*f1/(length(F)-1);
Cobg=b1'*g1/(length(G)-1);
Cobh=b1'*h1/(length(H)-1);
Cocd=c1'*d1/(length(D)-1);
Coce=c1'*e1/(length(E)-1);
Cocf=c1'*f1/(length(F)-1);
Cocg=c1'*g1/(length(G)-1);
Coch=c1'*h1/(length(H)-1);
Code=d1'*e1/(length(E)-1);
Codf=d1'*f1/(length(F)-1);
Codg=d1'*g1/(length(G)-1);
Codh=d1'*h1/(length(H)-1);
Coef=e1'*f1/(length(F)-1);
Coeg=e1'*g1/(length(G)-1);
Coeh=e1'*h1/(length(H)-1);
Cofg=f1'*g1/(length(G)-1);
Cofh=f1'*h1/(length(H)-1);
Cogh=g1'*h1/(length(H)-1);


%Putting all the calculated elements of covariance matrix we get
%Covariance matrix of size (8,8)
Com=[Coaa,Coab,Coac,Coad,Coae,Coaf,Coag,Coah;Coab,Cobb,Cobc,Cobd,Cobe,Cobf,Co
bg,Cobh;

Coac,Cobc,Cocc,Cocd,Coce,Cocf,Cocg,Coch;Coad,Cobd,Cocd,Codd,Code,Codf,Codg,Co
dh;

Coae,Cobe,Coce,Code,Coee,Coef,Coeg,Coeh;Coaf,Cobf,Cocf,Codf,Coef,Coff,Cofg,Co
fh;
```

```
Coag,Cobg,Cocg,Codg,Coeg,Cofg,Cogg,Cogh;Coah,Cobh,Coch,Codh,Coeh,Cofh,Cogh,Co
hh]
% Since all the non-diagonal elements are negative the data shows negative
% trend

%The third step is to find the Eigenvalue and Eigenvector
% Eigen Vector and Eigen Value
[T,U] = size(Com);
% Here m=n therefore the Covm i.e., Covariance matrix is square matrix.
x=sym('x');
% x is used as the unknown eigenvalue
I=eye(8);
% Here I is identity matrix(8,8)
%for calculating eigenvalues we solve the determinant of (x*I-A) where A is
%any square matrix and I is identity matrix of same size as A and x is
%unknown eigenvalue.
det(x*I-Com)
eigva=solve(det(x*I-Com))
eigvalues =double(eigva)
L=sort(eigvalues,'descend')
K=(L/sum(L)*100)


% To see how well the PCA model describes the data, we display the
eigenvalues
% accounted for by successive PCs in a pareto plot.
figure;
pareto(K);
xlabel('Principal Component')
ylabel('Variance')


% The 3 component model accounts for 75% of the variation in the data,
% therefore by examining PCs 1 2 and 3,we are getting a view of the process
% which is 75% as good as examining all the original variables.


% for eigenvector calculation
v_without_eig=zeros(U,U);
for idx=1:U;
    si=eigvalues(idx);
    V=(Com-si*eye(U));
    V_red=rref(V);
    vi=zeros(U,1);
    vi(U)=1;
    a=sym('a');
    b=sym('b');
    c=sym('c');
    d=sym('d');
    e=sym('e');
    p=sym('p');
    q=sym('q');
    r=sym('r');
    %[1,0,0,0,0,0,0,0,-0.2142]*[a,b,c,d,e,p,q,r,1]=a-0.2142=0 ->a=0.2142
```

```
    %[0,1,0,0,0,0,0,0,-0.0220]*[a,b,c,d,e,p,q,r,1]=b-0.0220=0 ->b=0.0220
    %[0,0,1,0,0,0,0,0,-0.9918]*[a,b,c,d,e,p,q,r,1]=c-0.9918=0 ->b=0.9918
    %[0,0,0,1,0,0,0,0,0.6233]*[a,b,c,d,e,p,q,r,1]=d+0.6233=0 ->b=-0.6233
    %[0,0,0,0,1,0,0,0,-0.7754]*[a,b,c,d,e,p,q,r,1]=e-0.7754=0 ->b=0.7754
    %[0,0,0,0,0,1,0,0,-0.4843]*[a,b,c,d,e,p,q,r,1]=p-0.4843=0 ->b=0.4843
    %[0,0,0,0,0,0,1,0,-0.2547]*[a,b,c,d,e,p,q,r,1]=q-0.2547=0 ->b=0.2547
    %vi=[0.2142;0.0220;0.9918;-0.6233;0.7754;0.4843;0.2547;1]
    vi=[V_red(1:(U-1),U);1];
    %%%MATLAB likes to normalize eigenvectors
    vi=vi/norm(vi);
    v_without_eig(:,idx)=vi;
end

% Eigenvalue and Eigenvector calculation using matlab function
[Z,L]=eig(Com)

%Arranging eigenvectors in descending order on the basis of respective
%eigenvalue
Eigenvector=[Z(:,8),Z(:,7),Z(:,6),Z(:,5),Z(:,4),Z(:,3),Z(:,2),Z(:,1)];


% Final Data Calculation
% where Y is Row Adjusted data and U Row Feature vector
Y=[A1,B1,C1,D1,E1,F1,G1,H1]'
U=Eigenvector'
P=U*Y
W=P'


% Create a biplot of the PCA Eigenvectors and Final data.
% We can use the PCA model to get a lot more insight into the behaviour of
the process,
% by plotting the PCA Eigenvectors and Final data on the same plot.

figure
hold on;
scatter(W(:,1), W(:,2), 5, 'bo');
scale = max((abs(W(:,1:2)))) ./ max((abs(Eigenvector(:,1:2))));
scatter(Eigenvector(:,1)*scale(1),      Eigenvector(:,2)*scale(2),      'ro',
'filled');
for i = 1:numvars
    line([0,Eigenvector(i,1)*scale(1)],...
        [0,Eigenvector(i,2)*scale(2)], 'Color', 'r');
    text(Eigenvector(i,1)*scale(1)-1,               Eigenvector(i,2)*scale(2),
variables{i});
end
title('Principal Component Biplot');
xlabel('PC 1');
ylabel('PC 2');
xlim([-10,20]);


% figure;
% scatter(W(:,1), W(:,2),'.');
% xlabel('\bf PCA1');ylabel('\bf PCA2')
```

```
figure
hold on;
scatter(W(:,1), W(:,3), 5, 'bo');
scale = max((abs(W(:,1:3)))) ./ max((abs(Eigenvector(:,1:3))));
scatter(Eigenvector(:,1)*scale(1),      Eigenvector(:,3)*scale(3),      'ro',
'filled');
for i = 1:numvars
    line([0,Eigenvector(i,1)*scale(1)],...
        [0,Eigenvector(i,3)*scale(3)], 'Color', 'r');
    text(Eigenvector(i,1)*scale(1)-1,               Eigenvector(i,3)*scale(3),
variables{i});
end
title('Principal Component Biplot');
xlabel('PC 1');
ylabel('PC 3');
xlim([-10,20]);

% figure;
% scatter(W(:,1), W(:,3),'.');
% xlabel('\bf PCA1');ylabel('\bf PCA2');
```

1.5 Results and Discussion

Using Matlab, we have done PCA for the data from Nasville Dome. The Data consists of two hundreds samples and eight variables out of two are isotope data and the rest are element data in weight percentage. To do PCA, we have to first standardize the original data because PCA produces erroneous results for different scaled dataest. In figure 6 we can see that the variables of original data are plotted with different units due to which the variability trend cannot be interpreted well. Therefore we should normalize the data using log transformation or any other method to bring all variables in same unit. Figure 7 shows the normalized data plotted but it is very tough to interpret that data in one graph.

To interpret the results, the first step is to determine how many principal components to examine, at least initially. Although PCA will return as many principal components as there are variables (eight), the point of PCA is to reduce dimensionality, so we will concentrate our initial interpretations on the largest principal components. To do this, we calculate the percent of total variance explained by each principal component, and make a bar plot of that. Therefore for interpreting the trend for data with huge number of variable in a single graph we can use Principal Component Analysis which stores >70 information of the original data. After standardizing the data we use it for principal component analysis and calculate the covariance matrix.

Covariance matrix

	d13C	d18O	Al	Ca	Fe	Mg	Mn	Si
d13C	1.0000	-0.1881	0.0948	-0.1226	0.0732	0.0280	0.2541	0.0559
d18O	-0.1881	1.0000	0.2276	-0.0397	-0.3430	0.5765	-0.4197	-0.0798
Al	0.0948	0.2276	1.0000	-0.3191	0.3667	0.5782	0.0360	0.8562
Ca	-0.1226	-0.0397	-0.3191	1.0000	-0.4663	-0.0595	0.0013	-0.3059
Fe	0.0732	-0.3430	0.3667	-0.4663	1.0000	-0.0378	0.2677	0.5808
Mg	0.0280	0.5765	0.5782	-0.0595	-0.0378	1.0000	-0.1286	0.2405
Mn	0.2541	-0.4197	0.0360	0.0013	0.2677	-0.1286	1.0000	0.1994
Si	0.0559	-0.0798	0.8562	-0.3059	0.5808	0.2405	0.1994	1.0000

Covariance matrix tells about the trend between variables and here d18O and Ca decreases with increasing d13C whereas all others increase with increasing d13C. d18O shows negative trend with Ca, Fe, Mn and Si and positive trend with Al and Mg. Al shows negative trend only with Ca. Fe shows negative trend with Mg and positive trend with Mn and Si. Mg shows negative trend with Mn and positive trend with Si.

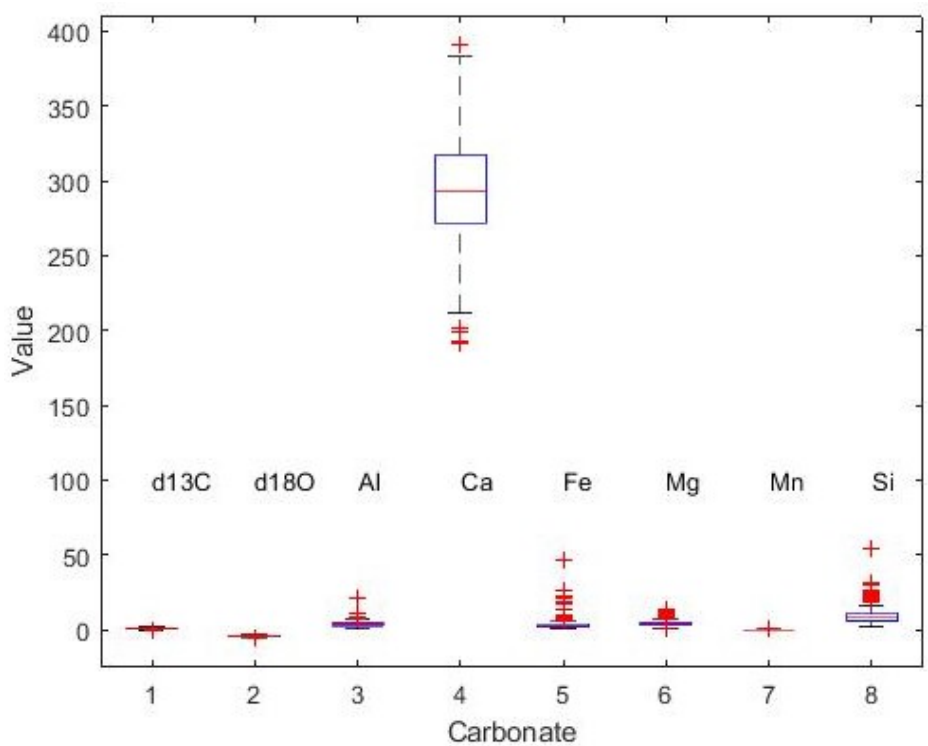

Figure 6. Plotting original data against Variables.

Eigenvectors

	PC8	PC7	PC6	PC5	PC4	PC3	PC2	PC1
d13C	-0.0404	0.1813	-0.0550	-0.3431	-0.5473	-0.6986	0.2084	0.1176
d18O	-0.0143	0.7011	0.1335	0.2921	-0.1469	-0.0384	-0.6180	0.0121
Al	0.7080	-0.1160	0.2503	-0.2025	0.1727	-0.0988	-0.2139	0.5444
Ca	0.0752	0.2839	-0.2792	-0.3520	0.6864	-0.3474	-0.0546	-0.3421
Fe	0.1330	0.3867	-0.6759	0.0485	-0.0309	0.3100	0.3089	0.4256
Mg	-0.2431	-0.4540	-0.5026	0.2356	0.0579	-0.3282	-0.4949	0.2658
Mn	0.0421	0.0861	0.1880	0.7032	0.2902	-0.4160	0.4292	0.1398
Si	-0.6424	0.1359	0.3062	-0.2823	0.2991	0.0564	0.0379	0.5489

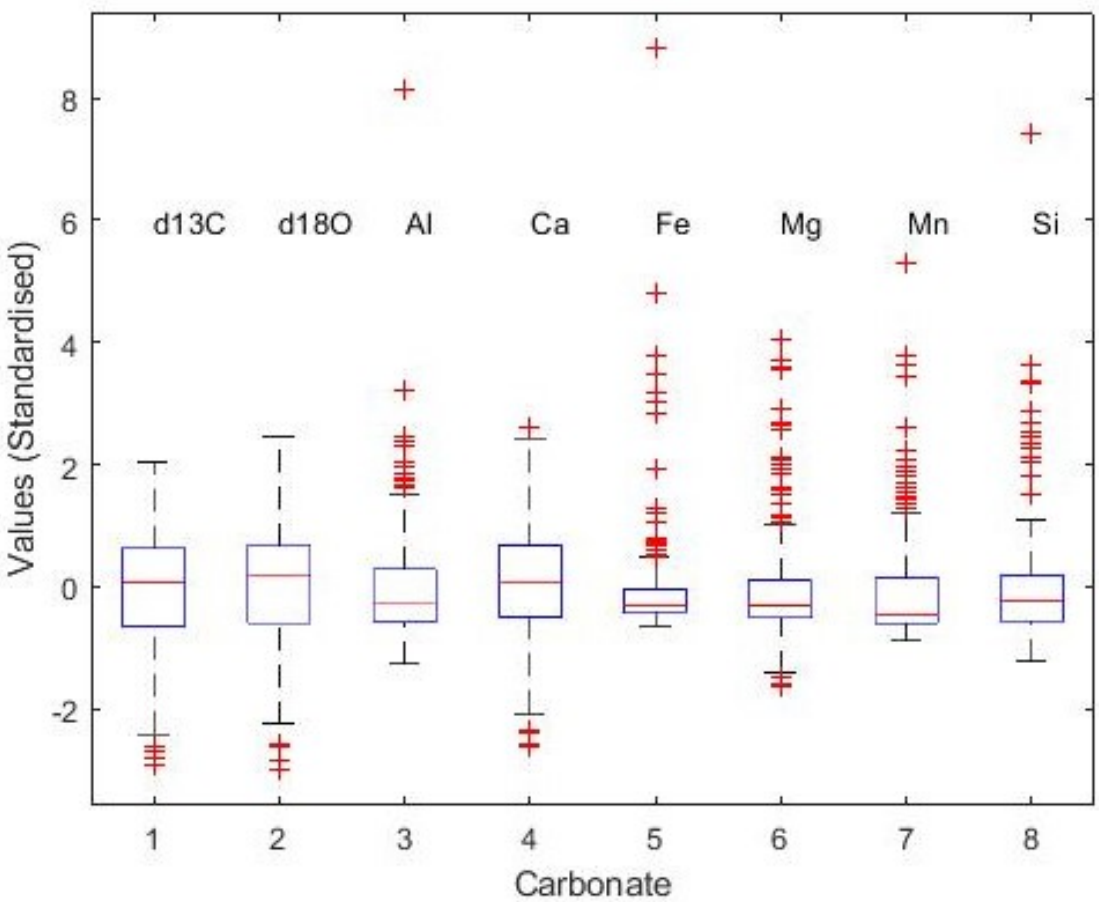

Figure 7. Plotting Standardized data against Variables.

Figure 8 shows the variance of different principal component which depicts that principal component 1, 2 and 3 represents more than 75 percent of our data, which suggest that we should

use these three principal components of higher significance in PCA and ignore other principal component of lesser significance.

Principal Components representing more than 75 percent information about our data

	PC1	PC2	PC3
d13C	0.1176	0.2084	-0.6986
d18O	0.0121	-0.6180	-0.0384
Al	0.5444	-0.2139	-0.0988
Ca	-0.3421	-0.0546	-0.3474
Fe	0.4256	0.3089	0.3100
Mg	0.2658	-0.4949	-0.3282
Mn	0.1398	0.4292	-0.4160
Si	0.5489	0.0379	0.0564

In our study from table shown above we can interpret that principal component 1 show positive trend for silicate and negative trend for calcium which indicates that the lithology was enriched in silica mineral and depleted in calcic minerals. Positive trend for iron, aluminum and magnesium indicates increasing clay contents. Positive trend for d13C, which may reflect oceanographic processes controlling the carbon isotopic composition of these samples. Principal component 2 show positive trend for silicate, manganese, iron and carbon isotopes and negative trend for calcium, magnesium, aluminum and oxygen isotopes. Principal component 3 show positive trend for silicate and iron and negative trend for all other. In this case, the first three principal components accounts for just over 75% of the explained variance.

The next step in our interpretation is to understand how our variables contribute to each of the principal components, and this is revealed by the loadings. Positive loadings indicate a variable and a principal component are positively correlated: an increase in one results in an increase in the other. Negative loadings indicate a negative correlation. Large (either positive or negative) loadings indicate that a variable has a strong effect on that principal component.

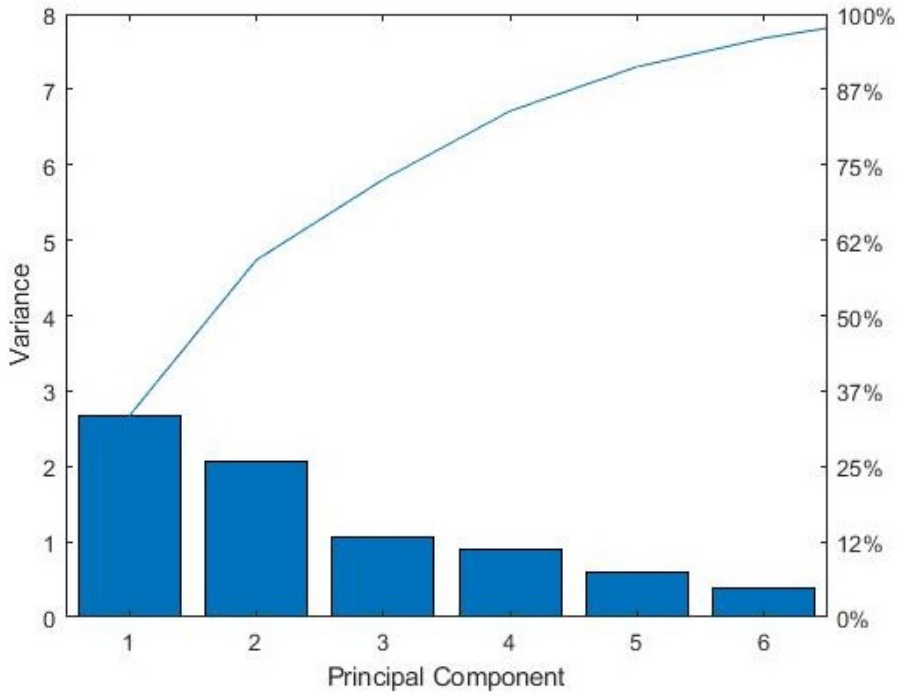

Figure 8. The percent of total variance explained by each principal component.

In this example, the loadings are readily interpretable. PC1 has a strong loading for calcium, silicon, iron, and aluminum. As these are analyses of limestones, this likely reflects relatively pure limestone at PC1 scores and increasing clay content. PC2 has strong loadings for d18O and magnesium, which may reflect dolomitization. PC3 is dominated by large loadings for d13C, which may reflect oceanographic processes controlling the carbon isotopic composition of these limestones.

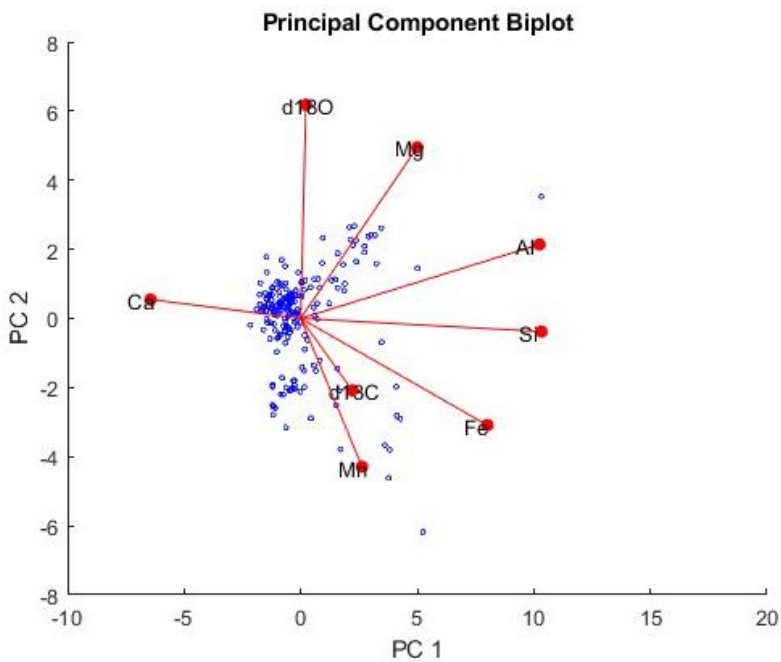

Figure 9. Plotting of Principal component 2 of the original data against Principal component 1.

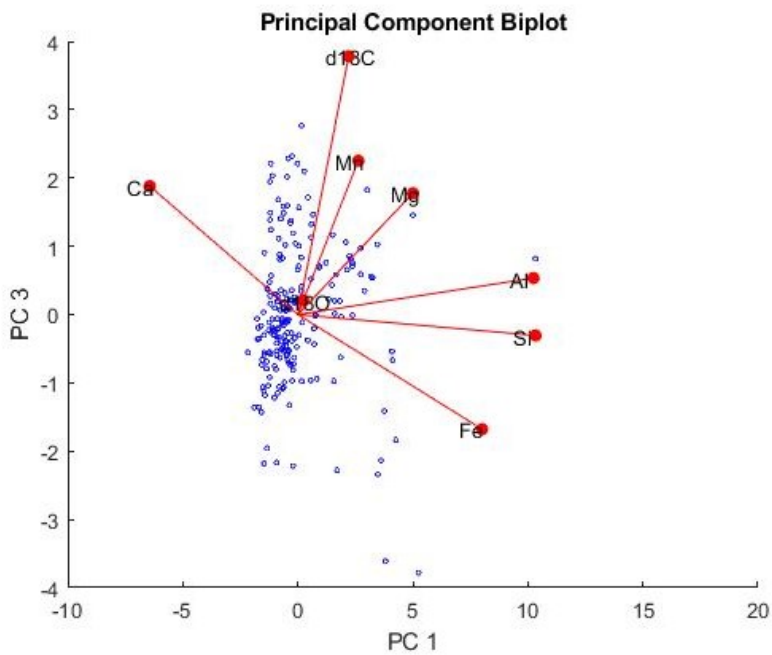

Figure 9. Plotting of Principal component 3 of the original data against Principal component 1.

Figure 9 and 10 shows the visual distribution of the samples using principal component 1 and 2 and principal component 1 and 3. From figure 9 we can interpret that the samples in first

principal component axis shows negative trend for calcium only and positive trend for all elements and isotopes whereas in second principal component axis it shows negative trend for oxygen isotopes and calcium.

1.5 Conclusion

In this study we carried out principal component analysis of eight dimensions data for 200 samples and from our study we make the following conclusion:

- We developed Matlab Algorithm for calculating PCA.
- We first applied PCA to a own made raw data which was about the marks for different subjects of nine students in a school.
- Then we applied PCA to data from Nashville dome, United States and the data comprises of 200 samples and eight dimensions.
- The samples analyzed were of different unit so they were first normalized and standardized so that all the variables make have equal effect on the analysis.
- We then calculated the principal components of the samples using several mathematical equations such as mean, covariance matrix, eigenvalue and eigenvector.
- We then identified the trends of the variables on the basis of principal components calculated which indicates that the samples analyzed were silica rich and depleted in calcic minerals and small positive trend in d13C indicates oceanographic processes controlling the carbon isotopic composition of these samples.

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